

Prediction

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$v_y = +2 \text{ m/s}$, $a_y = -10 \text{ m/s}^2$. Predict v_y one second later and explain.

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Acceleration tells change, not motion direction now.

H2 One second changes v_y by -10.

H3 Add +2 and -10.

CHECK AFTER ATTEMPT - PREDICTION

$v_y = -8 \text{ m/s}$.

The projectile is then moving downward.

Read the pattern

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

time 0,1,2,3 s; v_y +22,+12,+2,-8. Find a_y and when direction changes.

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Compare equal one-second intervals.

H2 Velocity drops by the same amount each second.

H3 Use $a = \Delta v / \Delta t$; direction changes at $v_y = 0$.

CHECK AFTER ATTEMPT - READ THE PATTERN

$a_y = -10 \text{ m/s}^2$.

Direction changes between 2 and 3 s.

Worked reconstruction

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

At $t=1$ s, $v_y=+12$, $a_y=-10$. Find u_y .

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Use one clock.

H2 Take upward positive.

H3 $12=u_y-10$.

CHECK AFTER ATTEMPT - WORKED RECONSTRUCTION

$u_y=22$ m/s upward.

Check: $22-10=12$.

Independent use

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

At $t=3$ s, $v_y=+3$, $a_y=-9$. Find u_y , state applicability, check.

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Recognise the event-clock structure.

H2 Use signed constant acceleration.

H3 $3=u_y-27$.

CHECK AFTER ATTEMPT - INDEPENDENT USE

$u_y=30$ m/s upward.

Model applies because vertical acceleration is constant and time is from launch.

Apex integrity

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

At the highest point, what are v_y and a_y , and why does the projectile fall afterward?

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Separate velocity from acceleration.

H2 Zero velocity at an instant does not imply zero acceleration.

H3 Gravity remains downward.

CHECK AFTER ATTEMPT - APEX INTEGRITY

At apex $v_y=0$ and $a_y=-g$.

Gravity drives v_y negative immediately afterward.